PREGUNTAS Y MODELOS

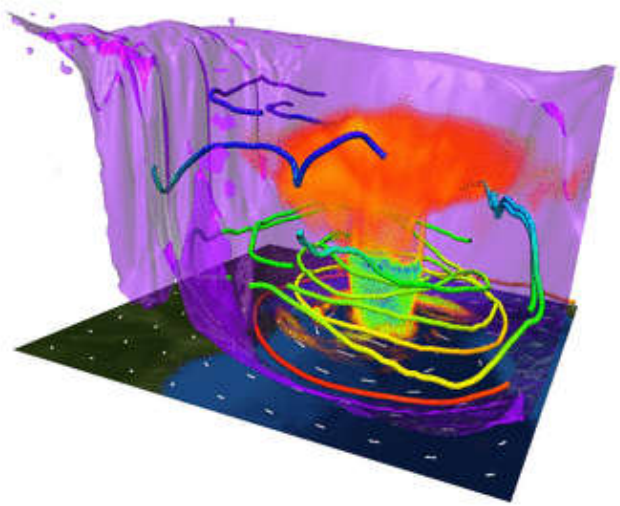
Yuley Cardona Orozco

¿QUÉ ES UN MODELO NUMÉRICO?

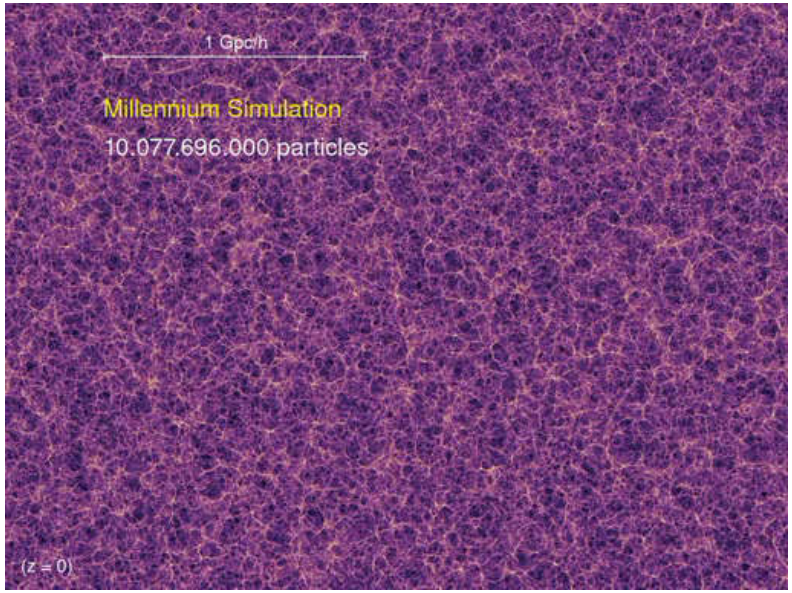
Es un abstracción de la realidad

¿POR QUÉ USAR MODELOS NUMÉRICOS?

Es una forma cuantitativa para entender y analizar sistemas complejos o fenómenos



A 3-D animated image of downscaled Global Forecast System (GFS) model data showing Hurricane Katrina making landfall on August 29, 2005. NOAA



The Millennium Simulation Project

The evolution of the matter distribution in a cubic region of the Universe

¿POR QUÉ USAR MODELOS NUMÉRICOS?

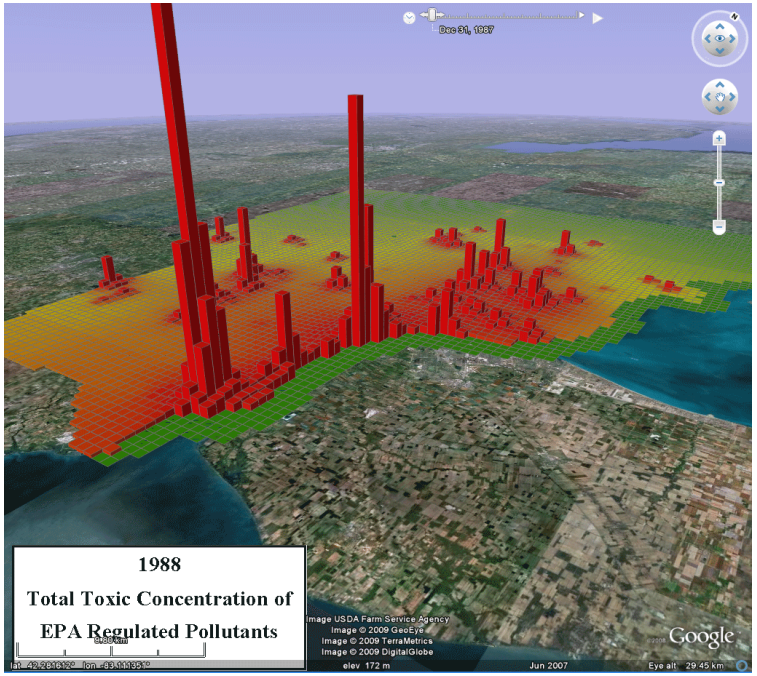
Para entender los procesos que dan forma al ambiente que nos rodea.

The erosion and deposition

http://www.cs.unc.edu/~blloyd/comp259/project/

¿POR QUÉ USAR MODELOS NUMÉRICOS?

Para visualizar escenarios de acción – Decisores políticos



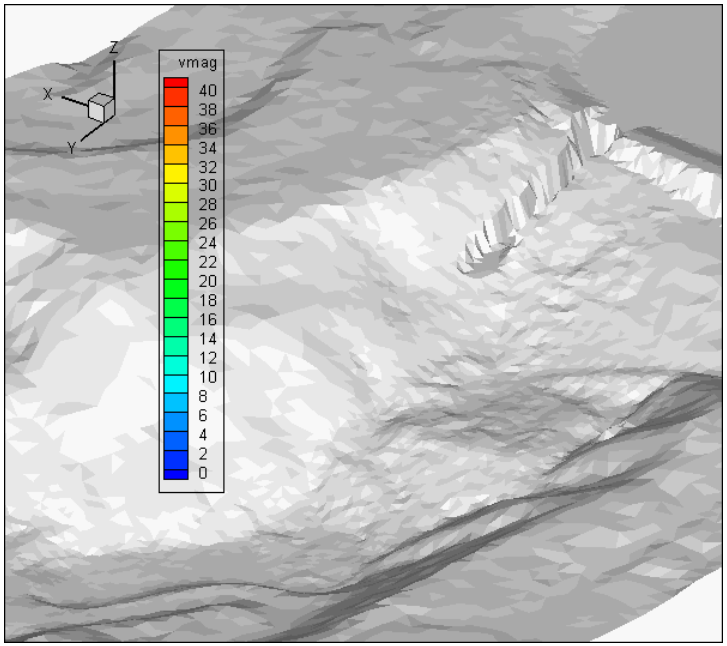
The toxicity of over 600 individual chemicals, amounts released, the path the chemical is expected to take as it is disbursed into the environment (determined by weather patterns Macomb Counties in Michigan for the years 1988- 2004.

EPA Measurements and Model

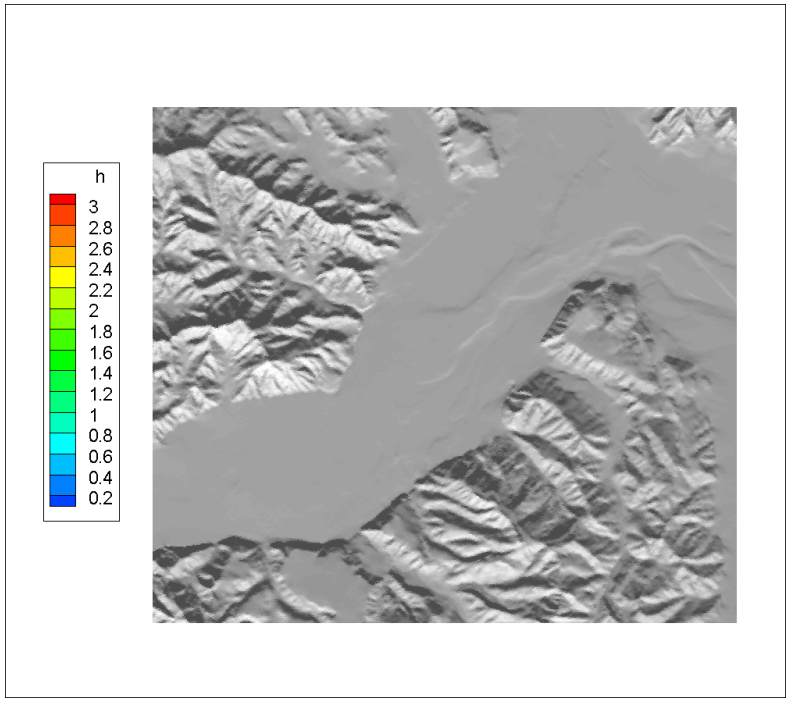
http://www-personal.umich.edu/~copyrght/image/solstice/sum09/Ard.html

¿POR QUÉ USAR MODELOS NUMÉRICOS?

Pueden ser útiles en el análisis de escenarios ¿Qué tal sí?



Baldwin Hills Dam-Break Flood

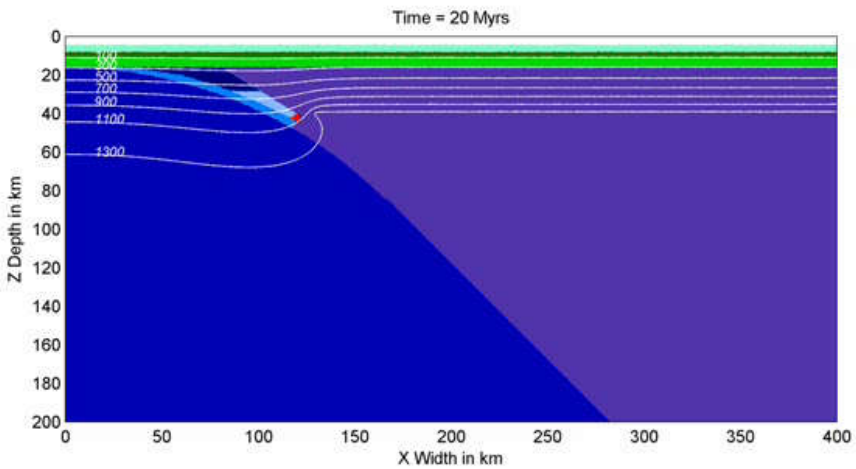


River basin flooding

Hydrodynamic flood simulation algorithm by *Brett F. Sanders* and *Lorenzo Begnudelli*

¿POR QUÉ USAR MODELOS NUMÉRICOS?

Algunas veces remplazan fases experimentales cuando estas pueden ser muy grandes, muy costosas muy perligrosas, o muy lentas.

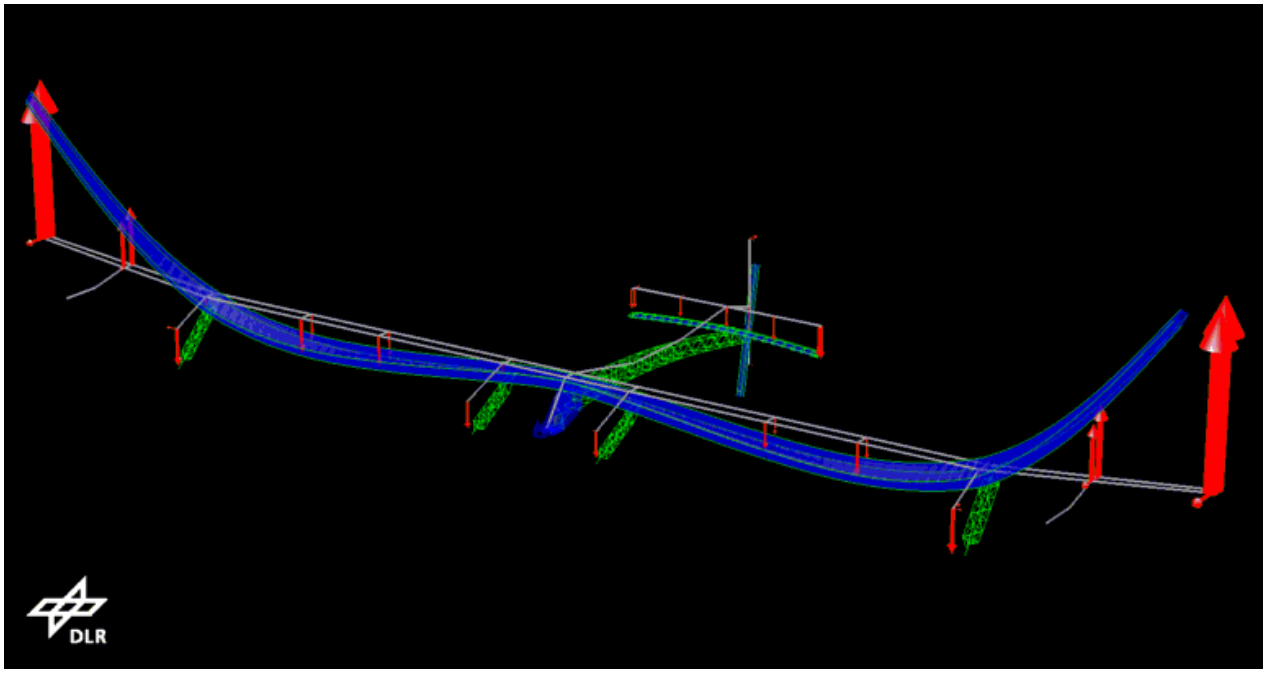


Rigid body rotation ("subduction wheel") in oceanic subduction (Gorczyk et al., 2006):
Lithology

http://www.erdw.ethz.ch/people/tgerya/Geodynamics

¿POR QUÉ USAR MODELOS NUMÉRICOS?

Algunas veces remplazan fases experimentales cuando estas pueden ser muy grandes, muy costosas muy perligrosas, o muy lentas.

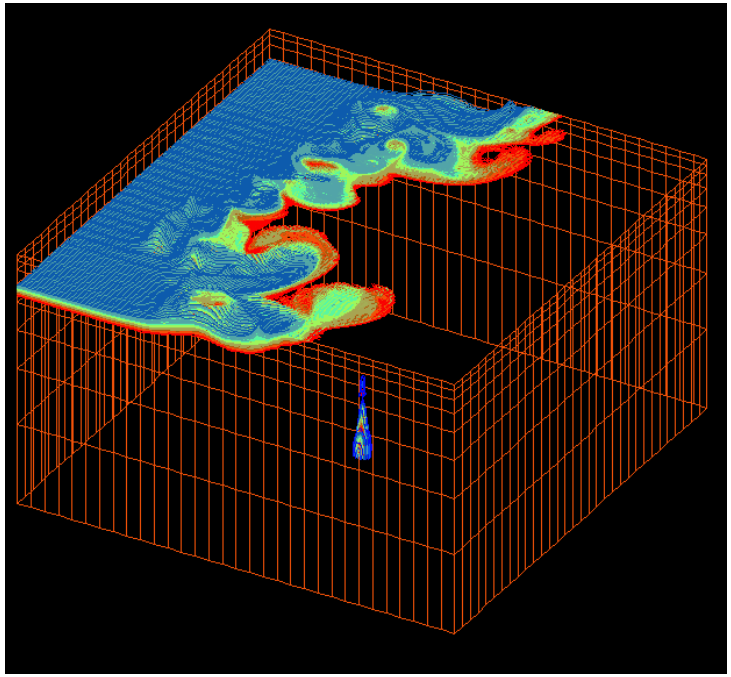


www.dlr.de

The animation shows wing deflection measured during the stationary vibration test

¿POR QUÉ USAR MODELOS NUMÉRICOS?

Algunas veces remplazan fases experimentales cuando estas pueden ser muy grandes, muy costosas muy perligrosas, o muy lentas.



3D buoyant plume released from the bottom of the ocean interacting with surface currents

http://www.rsmas.miami.edu/oilspill/models/

¿POR QUÉ USAR MODELOS NUMÉRICOS?

¿QUÉ MODELAR?

¿QUÉ ES LA MODELACIÓN AMBIENTAL?

Es una ciencia que combina matemática y computador para simular fenómenos físicos y químicos

Pasado

Katherine Johnson sits at her desk with a calculating machine and a special globe known as a celestial training device. Credit: NASA

Presente

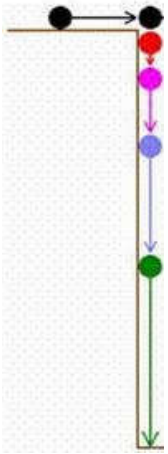
PREMISAS

El ambiente es algo que inicialmente puede parecer complicado.

- Los procesos no son observables, pero sus efectos y resultados si lo son.

- Los modelos numéricos son basados en modelos conceptuales.

SIMPLE → *COMPLEJO*



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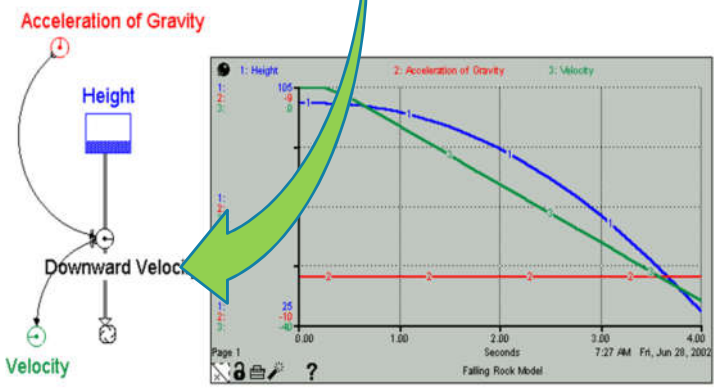
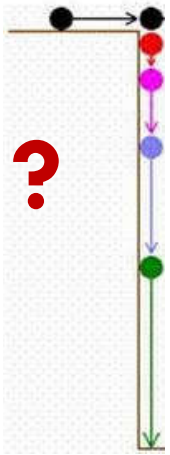
[redacted]



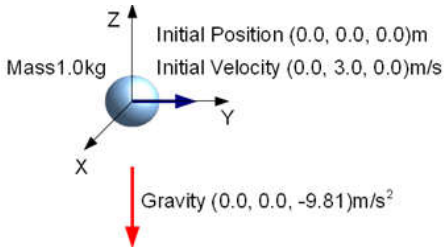
SIMPLE → COMPLEJO

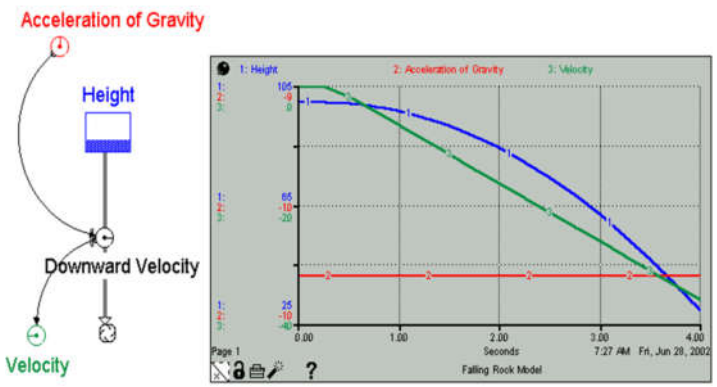
Caída Libre



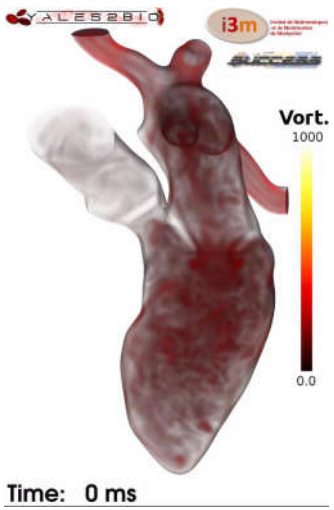
$$d = v * t$$

$$v = a * t$$





? Hemodynamics within a human left heart



Animation of the vorticity magnitude in a realistic human left heart. (from C. Chnafa, I3M)

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad \text{----- Continuity Equation} \quad (1)$$

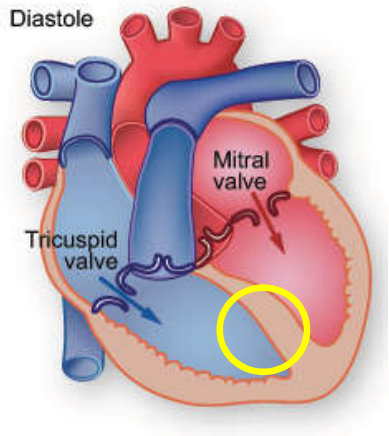
$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \mathbf{F} + \frac{\mu}{\rho}\nabla^2 \mathbf{u}, \quad \text{----- Equations of Motion} \quad (2)$$

$$\rho\left(\frac{\partial \varepsilon}{\partial t} + \mathbf{u} \cdot \nabla \varepsilon\right) - \nabla \cdot (K_H \nabla T) + p \nabla \cdot \mathbf{u} = 0. \quad \text{----- Conservation of Energy} \quad (3)$$

http://www.math.univ-montp2.fr/~nicoud/research.html

PEQUEÑA → GRAN ESCALA

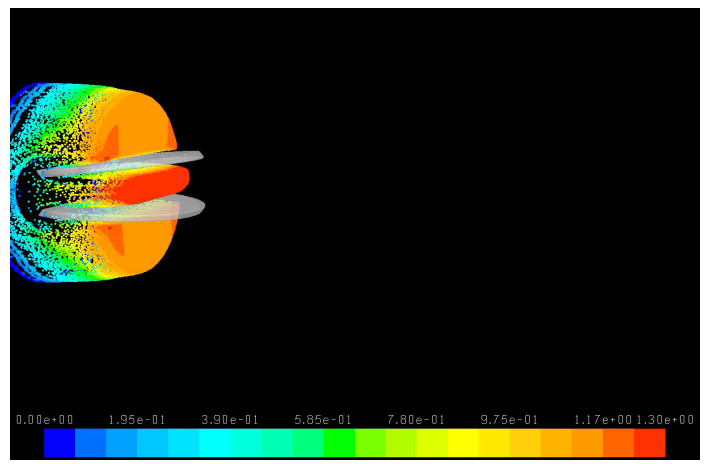
? **Blood flow past prosthetic heart valves**



$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_j}(\rho u_j k) = \rho P - \rho \omega k + \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right],$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_j}(\rho u_j \omega) = C_{\omega 1}\frac{\rho P \omega}{k} - C_{\omega 2}\rho \omega^2 + \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\omega}\right)\frac{\partial \omega}{\partial x_j}\right]$$

Modeling using **sophisticated non-Newtonian and turbulence models** to closely simulate the complex interactions between the blood constituents and the flow within vascular devices

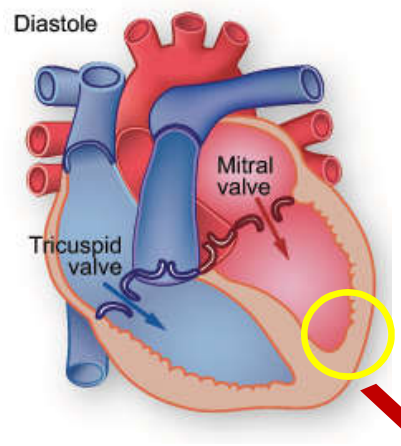


Blood flow past a St. Jude's bi-leaflet MHV (Mechanical Heart Valves) - Velocity vectors of particles are shown in this 3D picture.

http://bme.sunysb.edu/labs/dbluestein/cfd.html

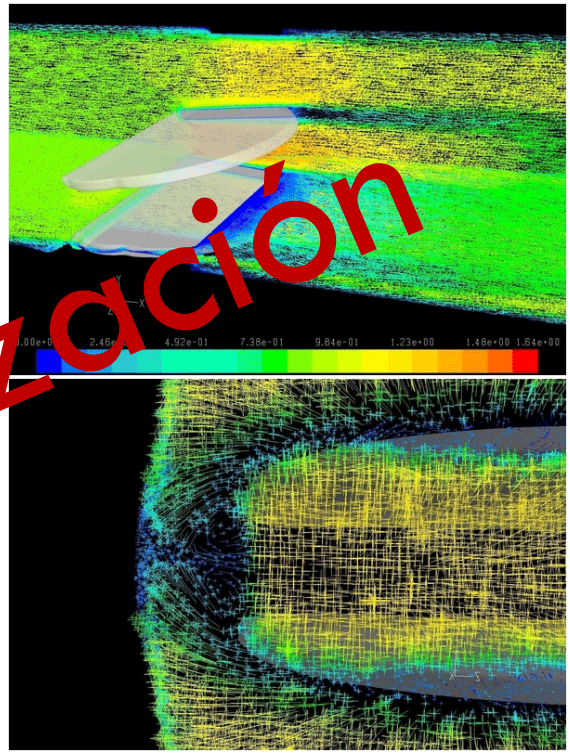
PEQUEÑA → GRAN ESCALA

? **Blood flow past prosthetic heart valves**



Visualización

Modeling using **sophisticated non-Newtonian and turbulence models** to closely simulate the complex interactions between the blood constituents and the flow within vascular devices



Animation of particles flowing past a St. Jude's MHV. The simulation is unsteady and turbulent with the fluid modeled as having same properties as blood.

http://bme.sunysb.edu/labs/dbluestein/cfd.html

BIOFLUIDS LABORATORY

The Biofluids Laboratory is one of the leaders in modeling complex blood flow past prosthetic heart valves using sophisticated non-Newtonian and turbulence models to closely simulate the complex interactions between the blood constituents and the flow within vascular devices, and in healthy and diseased conditions in the circulation.

Numerical Simulation/Modeling of Platelet Activation with CFD

Our **3D models** represent carefully obtained **accurate physiological geometries,** blood is modeled as a **non-Newtonian viscoelastic fluid** under **transient turbulent flow**conditions. To account for the transient nature of the flow past cardiovascular devices, we employ the innovative **Wilcox k-Omega** two-equations turbulence model:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_j}(\rho u_j k) = \rho P - \rho \omega k + \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right],$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_j}(\rho u_j \omega) = C_{\omega 1}\frac{\rho P \omega}{k} - C_{\omega 2}\rho \omega^2 + \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\omega}\right)\frac{\partial \omega}{\partial x_j}\right]$$

To simulate the interaction between the turbulent flow and the blood-borne particulates we employ **Stochastic models** (Gosman and Ioannides, 1981) that simulate the **influence of turbulence on particle (platelet) trajectories.** We further employ sophisticated damage accumulation models to characterize 'hot spot' areas in the flow field that may lead to thromboembolism.

PEQUEÑA → GRAN ESCALA

? Blood flow past prosthetic heart valves

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_j}(\rho u_j k) = \rho P - \rho \omega k + \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right],$$

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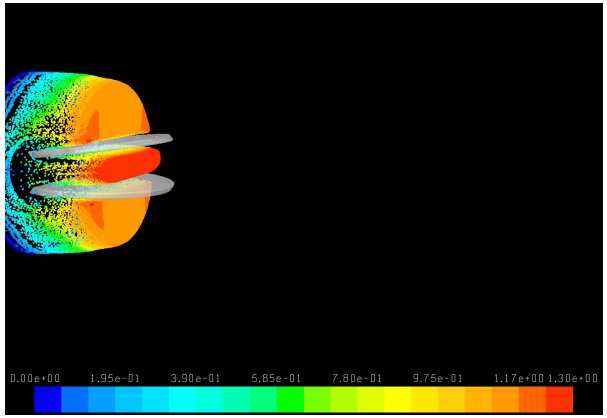


Blood flow past a St. Jude's bi-leaflet MHV (Mechanical Heart Valves) - Velocity vectors of particles are shown in this 3D picture.

http://bme.sunysb.edu/labs/dbluestein/cfd.html

? Shock waves around airplanes

Illustration of subsonic, transonic and supersonic velocity

K-epsilon turbulence model

Turbulent kinetic energy (k) and the Specific dissipation rate (ε).

Turbulent Kinetic Energy (TKE) Equation:

$$\rho\frac{\partial K}{\partial t} + \rho U\frac{\partial K}{\partial x} + \rho V\frac{\partial K}{\partial y} + \rho W\frac{\partial K}{\partial z}$$
$$= \frac{\partial}{\partial x}\left[\left(\frac{\mu_t}{\sigma_K}\right)\frac{\partial K}{\partial x}\right] + \frac{\partial}{\partial y}\left[\left(\frac{\mu_t}{\sigma_K}\right)\frac{\partial K}{\partial y}\right] + \frac{\partial}{\partial z}\left[\left(\frac{\mu_t}{\sigma_K}\right)\frac{\partial K}{\partial z}\right] - \rho\varepsilon +$$
$$\mu_t\left[2\left(\frac{\partial U}{\partial x}\right)^2 + 2\left(\frac{\partial V}{\partial y}\right)^2 + 2\left(\frac{\partial W}{\partial z}\right)^2 + \left(\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x}\right)^2 + \left(\frac{\partial U}{\partial z} + \frac{\partial W}{\partial x}\right)^2 + \left(\frac{\partial V}{\partial z} + \frac{\partial W}{\partial y}\right)^2\right]$$

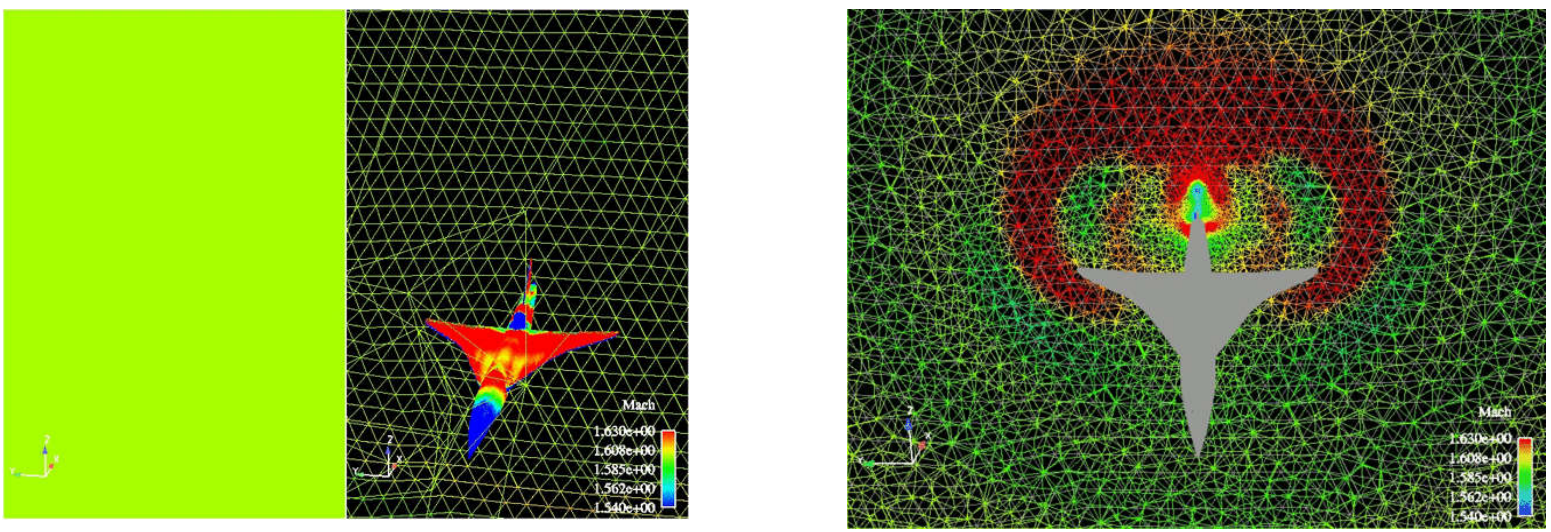
Turbulent Energy Dissipation (TED) Equation:

$$\rho\frac{\partial \varepsilon}{\partial t} + \rho U\frac{\partial \varepsilon}{\partial x} + \rho V\frac{\partial \varepsilon}{\partial y} + \rho W\frac{\partial \varepsilon}{\partial z}$$
$$= \frac{\partial}{\partial x}\left[\left(\frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x}\right] + \frac{\partial}{\partial y}\left[\left(\frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial y}\right] + \frac{\partial}{\partial z}\left[\left(\frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial z}\right] - C_2\rho\frac{\varepsilon^2}{K}$$
$$C_1\mu_t\frac{\varepsilon}{K}\left[2\left(\frac{\partial U}{\partial x}\right)^2 + 2\left(\frac{\partial V}{\partial y}\right)^2 + 2\left(\frac{\partial W}{\partial z}\right)^2 + \left(\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x}\right)^2 + \left(\frac{\partial U}{\partial z} + \frac{\partial W}{\partial x}\right)^2 + \left(\frac{\partial V}{\partial z} + \frac{\partial W}{\partial y}\right)^2\right]$$

Pequeña → Gran escala

? Shock waves around airplanes

Illustration of subsonic, transonic and supersonic velocity



Computed shock waves

http://asn.epfl.ch/Aerodynamics

Flows around planes

A promising application for mesh adaptation is the computation of compressible flows around aircrafts. This is a collaboration with Dassault Aviation, INRIA (F. Alauzet, Gamma project) and Y. Bourgault (Univ. Ottawa).
During flight, an airplane moves the surronding air in a similar way that a stone thrown into water. The generated waves propagate in every direction with the velocity of the sound (acoustic waves). When a plane travells at a velocity close to the sound velocity (transonic speed), the waves accumulate at the front of the plane increasing the resistivity of the air (sonic barrier). If the velocity increase again, it becomes supersonic and shock waves can appears on the entire structure of the plane.

n the later case, the shock waves generate huge stress on the structure and thus need to be predicted correctly. To do that the chair is using mesh adaptation techniques that largely improve the precision of the computed shock waves as illustrated in the animations below

CORTO TIEMPO → LARGO TIEMPO

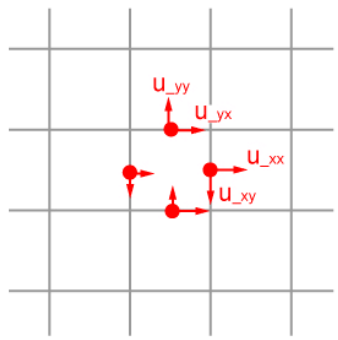
Simulation of pyroclastic flows

? The rheological behaviour of pyroclastic flows

$$\frac{\partial h}{dt}+\frac{\partial}{\partial x}(hu_x)+\frac{\partial}{\partial y}(hu_y)=0$$

$$\frac{\partial}{\partial t}(hu_x)+\frac{\partial}{\partial x}(hu_x^2)+\frac{\partial}{\partial y}(hu_xu_y)=gh\sin\alpha_x-\frac{1}{2}k_{actpass}\frac{\partial}{\partial x}(gh^2\cos\alpha)+\frac{T_x}{\rho}$$

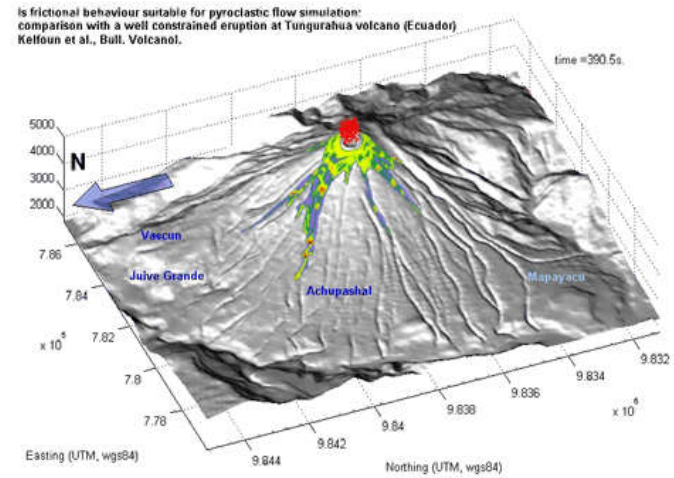
$$\frac{\partial}{\partial t}(hu_y)+\frac{\partial}{\partial x}(hu_y^2)+\frac{\partial}{\partial y}(hu_yu_x)=gh\sin\alpha_y-\frac{1}{2}k_{actpass}\frac{\partial}{\partial y}(gh^2\cos\alpha)+\frac{T_y}{\rho}$$



1. For each time increment VolcFlow first computes the velocity at the edges with the source-terms only The fluxes of mass and momentum are then calculated using a first upwind scheme.

2. The second stage of the algorithm computes the advection terms and calculates the new mean velocity at t

3. Finally, the x an the new velocities modified by advec using a second up



http://wwwobs.univ-bpclermont.fr/lmv/perso/Kelfoun_Karim/VolcFlow/VolcFlow.html

CORTO TIEMPO → LARGO TIEMPO

Earth's largest environmental catastrophe 250 million years ago

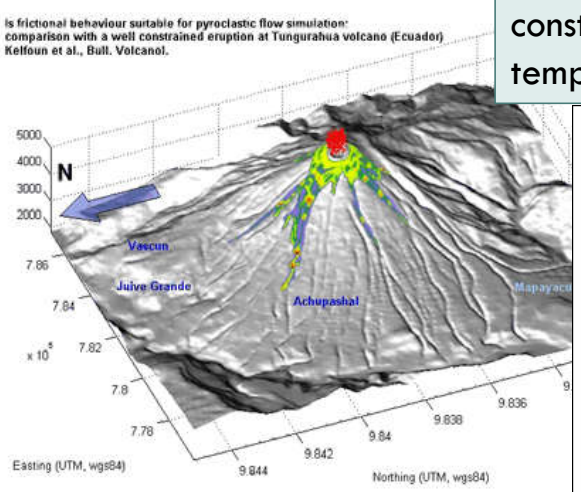
Simulation of pyroclastic flows

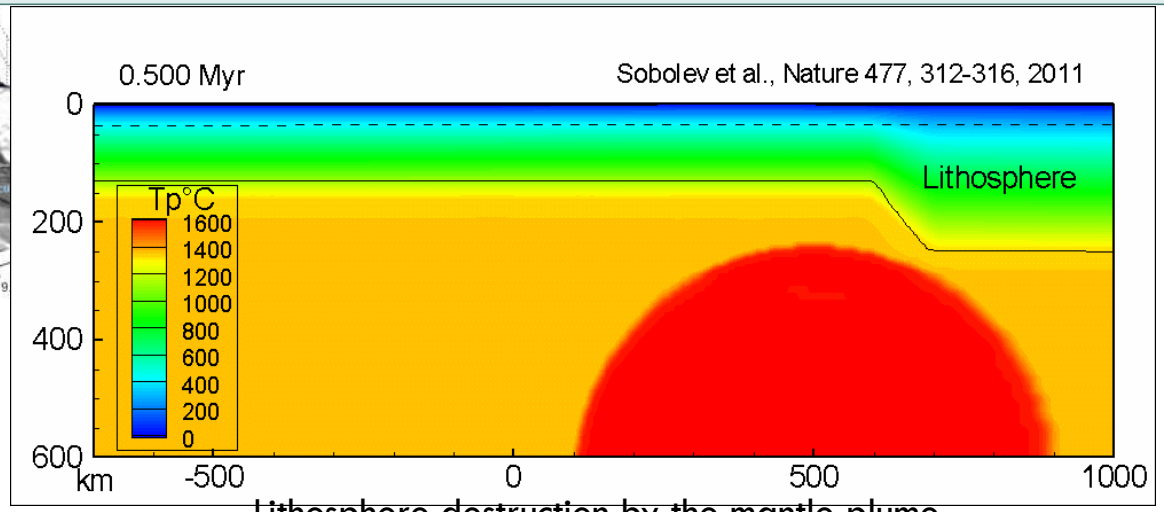
The eruption of giant masses of magma in Siberia 250 million years ago led to the Permo-Triassic mass extinction when more than 90 % of all species became extinct.

? The rheological behaviour of pyroclastic flows

Thermomechanical model

The model assumes source-composition and temperature based on petrological constraints, employs non-linear elasto-visco-plastic rheology and pressure- and temperature dependent melting of a heterogeneous mantle.





Lithosphere destruction by the mantle plume

http://wwwobs.univ-bpclermont.fr/lmv/perso/Kelfoun_Karim/VolcFlow/VolcFlow.html

http://phys.org/news/2011-09-earth-largest-environmental-catastrophe-million.html#jCp

MENOS ES MAS!

PARSIMONIA

¿Cuál es el modelo óptimo?

Es aquel que contiene la complejidad suficiente para explicar un fenómeno

Principio de Parsimonia: Implementar un modelo que solo contenga la complejidad que es absolutamente necesaria para representar la realidad.

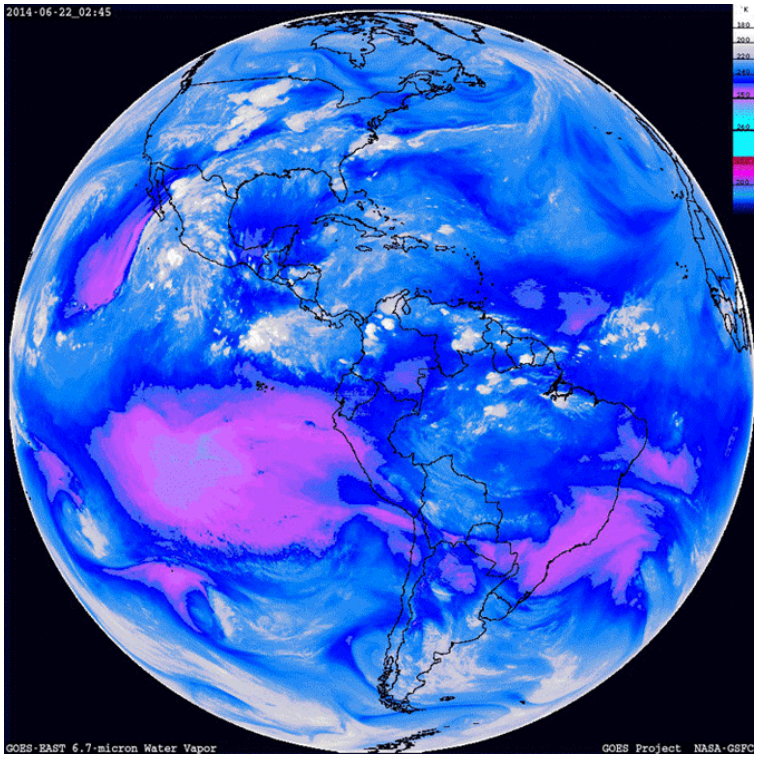
Principio de Newton para la investigación "We are to admit no more causes of natural things than such as are both true and sufficient to explain their appearances"

NUNCA OLVIDAR

Un modelo **no** validado no es mejor que una hipótesis no probada

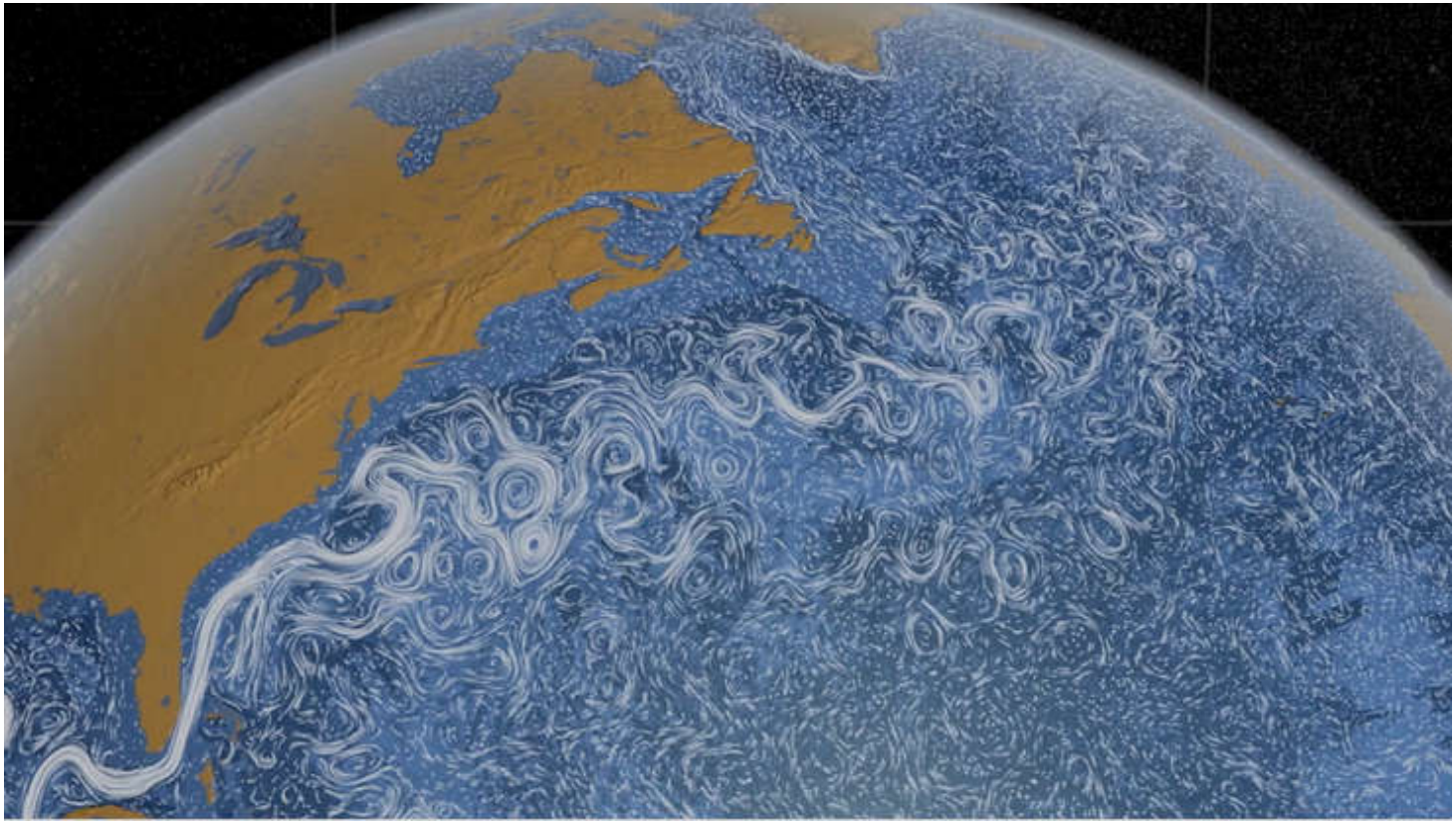
- Modelar **no** es una alternativa para la observación, pero bajo algunas circunstancias puede ser una herramienta poderosa para entender las observaciones y probar teorías.

- Las observaciones serán siempre **más** cercanas a la realidad y es la componente más importante de la investigación científica



The movement of upper-air water vapor over the Eastern Pacific is shown using GOES satellite air temperature data. High, cold clouds are white. High, cold, clear air (around -28 F) is blue. Lower, warmer, dry air (around -10 F) is magenta (where clear, dry air penetrates lower in the atmosphere). Credit: NASA/NOAA GOES Project Dennis Chesters

Read more at: http://phys.org/news/2014-06-nasanoaa-vapor-animations-oceans.html#jCp



NASA animation shows the global movement of Earth's ocean surface currents from June 2005 to December 2007. The animation was created using data from NASA satellites, direct ocean measurements and a numerical model developed by NASA JPL and the Massachusetts Institute of Technology. Credit: NASA/SVS